

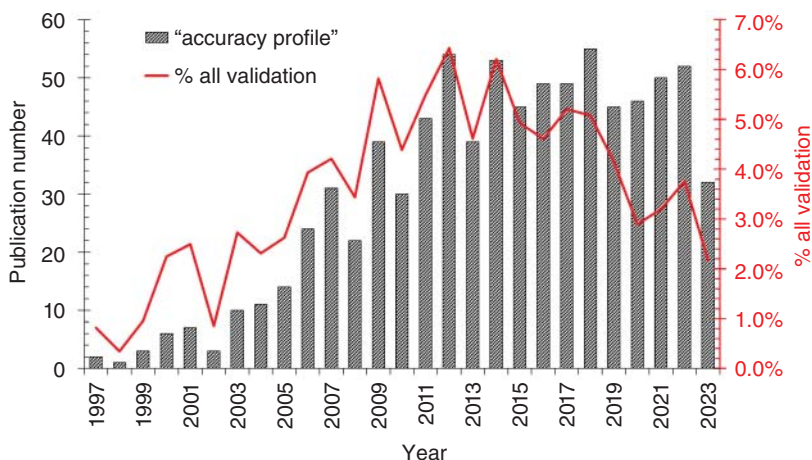


Figure 5.4 Number of publications with “accuracy profile” in the title ratioed to all “validation” published papers.

where $\bar{Z}$ is the average inverse-predicted concentration of replicates of the validation material, k_{TI} the coverage factor of the interval depending on the chosen probability β , and s_{TI} the standard deviation, which will be defined further. At each concentration level, i.e. for each validation material, the β -ETI bounds are separately computed and are graphically connected to draw a profile, giving its name to the procedure.

It has been recognized that the performance of a method is greatly dependent on the concentration in a nonlinear way (see Section 7.5). Therefore, the linear interpolation between interval bounds is incorrect, but it is a simple and convenient solution to illustrate and graphically interpret a MAP. But the use of straight-line segments when profiles are expressed as percentages to the reference concentration can be misleading for some complementary calculations as explained in the next section about the limit of quantification.

Since the first publication in 1996, the number of validation examples based on the MAP has increased. Figure 5.4 illustrates the results of a simple query of the document database Science Direct, run by Elsevier. Searched papers must contain the term “accuracy profile” in the title. About 815 publications were found, but it can be assumed that this number is an underestimation since this validation procedure was standardized and may have been used but not cited in the title, or unpublished analytical methods were validated using an accuracy profile.

Over the same period, the terms “method validation” and “analytical method” appeared in approximately 21,000 publications in chemistry journals. Thus, 2% and 5% of the publications were based on the MAP procedure. This significant difference is due to semantic ambiguity. Many analysts use the term “validation” when developing or characterizing a new method and just checking the specificity, i.e. verifying the method is correctly determining the analyte it is supposed to.

Table 5.1 THEOPHYLLINE – description of the dataset.

Title of the dataset	THEOPHYLLINE
Reference publication	[23]
Measurand	Theophylline concentration in plasma, expressed in $\mu\text{g/l}$
Measurement method	Ultra-High Pressure Liquid Chromatography (UHPLC) coupled with a tandem mass spectrometer detection UHPLC-MS/MS
Validation range	[0.05, 10] $\mu\text{g/l}$
Acceptance interval	$\pm 25\%$. Classic value for biological analysis
Validation materials	Six validation materials containing 0.05, 0.1, 0.5, 1.0, 2.5, and 10.0 $\mu\text{g/l}$ respectively, prepared by SAM from a batch of homogenized plasma
Validation design	Series ($I = 6$), replicates/series ($J = 2$), levels ($K = 6$)
Calibration design	Series ($I = 6$), replicates/series ($J' = 2$), levels ($K' = 5$) containing 0.02, 0.1, 0.5, 2.5, and 10.0 $\mu\text{g/l}$ respectively
Total number of measures	60 measurements on calibration solutions. 72 measurements on spiked materials.
Predicted concentrations	Table 5.2
Statistics	Table 5.3

5.2.2 Method Accuracy Profile by Example

The THEOPHYLLINE dataset already used in Section 2.2 will illustrate the steps described above in building the MAP and explain the final interpretation. Details on instrumental conditions and sample preparation are described in the original publication [23]. The dataset features are summarized in Table 5.1. Calibration data are not presented, except for the series 1 data already put together in Table 1.3 to illustrate the different applicable algorithms to estimate a calibration curve. As explained, different calibration models and regression methods apply to the same dataset to obtain different calibration curves and, therefore, inverse-predicted concentrations. For all series, the latter expressed in $\mu\text{g/l}$, are gathered in Table 5.2. They were calculated using second-order polynomials fitted by WLS regression. In Section 8.1, the role of the different regression techniques will be illustrated to demonstrate how the accuracy profile is a very efficient tool for selecting the best calibration function.

All details on the formulas used to obtain inverse-predicted concentrations are already described in Section 5.3.1. Compared to the original publication [23], the results in Table 5.2 are rounded to facilitate a possible use as benchmark. This remark is intended to explain slight differences between the calculated data and those originally published. Because many formulas use sums of squares, an even small difference may result in an even more important difference in the final result. In Table 5.3, we gather the basic statistical parameters that are necessary to draw

Table 5.2 THEOPHYLLINE – inverse-predicted concentrations of the validation materials obtained by WLS quadratic regression ($\mu\text{g/l}$).

Calibrator ($\mu\text{g/l}$)	Replicate	Series					
		1	2	3	4	5	6
0.05	1	0.077	0.052	0.055	0.049	0.051	0.076
	2	0.074	0.058	0.056	0.049	0.052	0.055
0.1	1	0.114	0.112	0.104	0.100	0.113	0.147
	2	0.113	0.110	0.101	0.105	0.107	0.112
0.5	1	0.534	0.509	0.479	0.593	0.538	0.506
	2	0.543	0.494	0.478	0.535	0.512	0.514
1.0	1	1.144	1.028	0.902	0.988	0.977	0.975
	2	1.113	0.996	0.892	1.074	0.957	0.970
2.5	1	2.560	2.372	3.127	2.888	2.380	2.420
	2	2.486	2.233	2.280	2.585	2.394	2.472
10	1	10.424	10.164	9.928	10.037	10.134	10.470
	2	10.829	10.606	9.286	10.832	10.518	10.998

a MAP. In this example, $\beta\%$ is set equal to 80%, which means that each β -ETI will contain, on average, 80% of the future measurements that could be obtained on these validation materials. For example at the concentration level of $0.5 \mu\text{g/l}$, 80% of possible measurement values are expected to lie between $[0.470, 0.569] \mu\text{g/l}$. Explanation and computation details are given further in Section 5.3.1.

The results summarized in Table 5.3 were obtained using the worksheet program called Resource H β -ETI (Excel) This remark is intended to explain that they can easily be reproduced by any analyst. The interval bounds are not directly used to represent the MAP graphically. Beforehand, they are ratioed to the reference value X assigned to each validation material. This convenient mode of expression is a percentage that can be interpreted as a recovery yield bounded with its dispersion interval β -ETI.

The relative values of TI limits must not be confused with the relative standard deviation (RSD) classically used by analysts to express method performances as described in Section 3.2.1. For instance, at level $X = 0.050 \mu\text{g/l}$, the average inverse-predicted concentration is $\bar{Z} = 0.059 \mu\text{g/l}$, and the standard deviation of repeatability $s_r = 0.0064 \mu\text{g/l}$. Thus, the RSD of repeatability for this level is $RSD_r = 100 \times \frac{0.0064}{0.059} = 10.8\%$. While the recovery yield interval at the same concentration is completely different $[84\%; 150\%]$. RSD is not interesting for the present objective of validation because it is not compared to an acceptance criterion. If $\bar{Z}$ was used instead of X to compute the relative TI bounds, the biases would be included and may modify the resulting values. Ratioing the intervals is also a way to illustrate the method profile in a familiar manner for analysts.

Table 5.3 THEOPHYLLINE – main validation parameters and β -expectation tolerance interval bounds.

Parameters		Levels					
Concentration ($\mu\text{g/l}$)	X	0.050	0.10	0.50	1.0	2.5	10.0
Lower acceptance limit	A_L	0.038	0.075	0.375	0.750	1.875	7.500
Upper acceptance limit	A_U	0.063	0.125	0.625	1.250	3.125	12.500
Precision parameters							
Number of series	I	6	6	6	6	6	6
Number of measures	$I \times J$	12	12	12	12	12	12
Number of replicates	J	2	2	2	2	2	2
Predicted concentration	$\bar{Z}$	0.059	0.112	0.520	1.001	2.516	10.352
Recovery yield	%	117	112	104	100	101	104
Repeatability std. dev.	s_r	0.0064	0.0104	0.0192	0.0287	0.2641	0.3905
Between-series std. dev.	s_B	0.0089	0.0067	0.0266	0.0748	0.0000	0.2841
Intermediate Precision	s_{IP}	0.0110	0.0124	0.0328	0.0802	0.2641	0.4829
β -Expectation Tolerance Intervals (β -ETI)							
Tolerance interval std. dev.	s_{TI}	0.0117	0.0130	0.0350	0.0862	0.2749	0.5093
Degrees of freedom	N_E	7.01	9.59	7.02	5.69	10.91	9.22
Coverage factor	k_{TI}	1.41	1.38	1.41	1.45	1.36	1.38
Proportion	$\beta\%$	80%	80%	80%	80%	80%	80%
Lower bounds		0.042	0.094	0.470	0.876	2.141	9.649
Upper bounds		0.075	0.129	0.569	1.126	2.891	11.055

Notations are explained in Section 5.3.1.

Figure 5.5 is based on Table 5.4 data and provides a typical example of an accuracy profile. Since the validation range is widely scattered, covering six concentration levels and three orders of magnitude between 0.05 and 10.0 $\mu\text{g/l}$, the *funnel* shape of the profile is visible. For the above-explained reasons, relative TIs are not symmetrical in respect to the 100% recovery line. The elements used to draw the MAP are as follows:

On the horizontal axis:

- The theoretical concentration of the analyte, denoted X , assigned to the validation materials and obtained, in this example, by standard additions.

On the vertical axis:

- The average recovery yield is the gray dots.
- The two bounds of the β -ETI of inverse-predicted concentrations Z , expressed as recovery yields and connected by interpolation solid lines.
- The limits of the acceptance intervals are two horizontal dashed lines, defined according to the objective of the method, also expressed in %, like the β -ETI bounds.

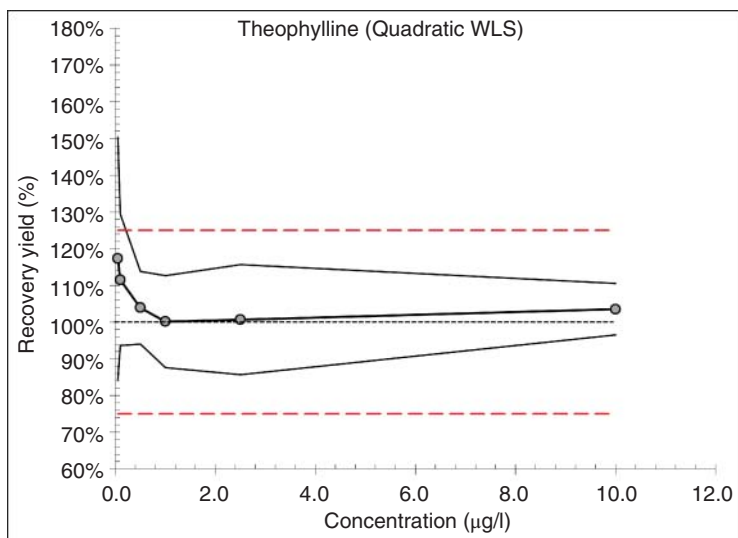


Figure 5.5 THEOPHYLLINE – MAP with six validation materials. Inverse-predicted concentrations are obtained with WLS quadratic model. Dots: average recovery yields. Solid lines: β -ETI for $\beta = 80\%$. Dashed lines: acceptance intervals ($\pm 25\%$).

Table 5.4 THEOPHYLLINE – β -expectation tolerance intervals relative to the reference values of the corresponding level.

Levels ($\mu\text{g/l}$)	0.050 (%)	0.10 (%)	0.50 (%)	1.0 (%)	2.5 (%)	10.0 (%)
Recovery yield (%)	117	112	104	100	101	104
Expectation tolerance interval $\beta = 0.80$	84	94	94	88	86	96
Acceptance interval	150	129	114	113	116	111
	75	75	75	75	75	75
	125	125	125	125	125	125

The graphical interpretation rules are simple and summarized in Figure 5.6:

1. When the bounds of the β -ETIs are within the acceptance interval, the method is said to be valid at this level of concentration because it can be predicted that at least $\beta\%$ of future measures will lie inside the acceptance interval, for instance at $\pm 25\%$ around the analyte assigned true value in this study.
2. When a single bound is outside the acceptance interval, the method is no longer considered capable of providing the expected proportion of accurate measurements.

It is possible to define the validated range of the method in more statistical terms. It is the concentration range where at least 80% of the measurements the method provides are considered acceptable. In Section 5.4.6, it is explained how to compute the proportion of results exceeding the bounds of the β -ETIs and defined as nonacceptable.